

Two Ways to Build an Interval

Unit 3 · Topic 3.10 · TODAY'S PROBLEM

Town Oak has 2,200 households and Pine 1,800. Independent SRSs find 77 of 140 Oak households and 48 of 120 Pine households use curbside composting. Let $p_O - p_P$ mean Oak minus Pine.

Elena: “The two-sample z -interval is $(0.030, 0.270)$.”

Devon: “Subtract the farthest endpoints of separate 95% intervals. That gives $(-0.020, 0.320)$, so zero is plausible.”

Household samples

Town	Use	Other	n	N
Oak	77	63	140	2200
Pine	48	72	120	1800

FIRST TAKE (5 min, on your sheet)

The two reported ranges disagree about whether the towns could be equal. Which range would you trust, and why?

- DOK 1** (a) Define p_O and p_P and name the interval procedure for $p_O - p_P$.
- DOK 2** (b) Check independent samples, both 10 percent conditions, and all four counts. Compute the estimate, SE, margin of error, and 95% interval.
- DOK 3** (c) In 3–4 sentences, choose a method using the independent samples and the variance formula. Explain why subtracting separate 95% intervals does not establish a 95% interval for the difference. Use zero's location to explain how the method changes the conclusion.

Optional review: [follow-along](#) (scan the code)

Work (a)–(c) from the rules on your sheet. Turn in your sheet.

